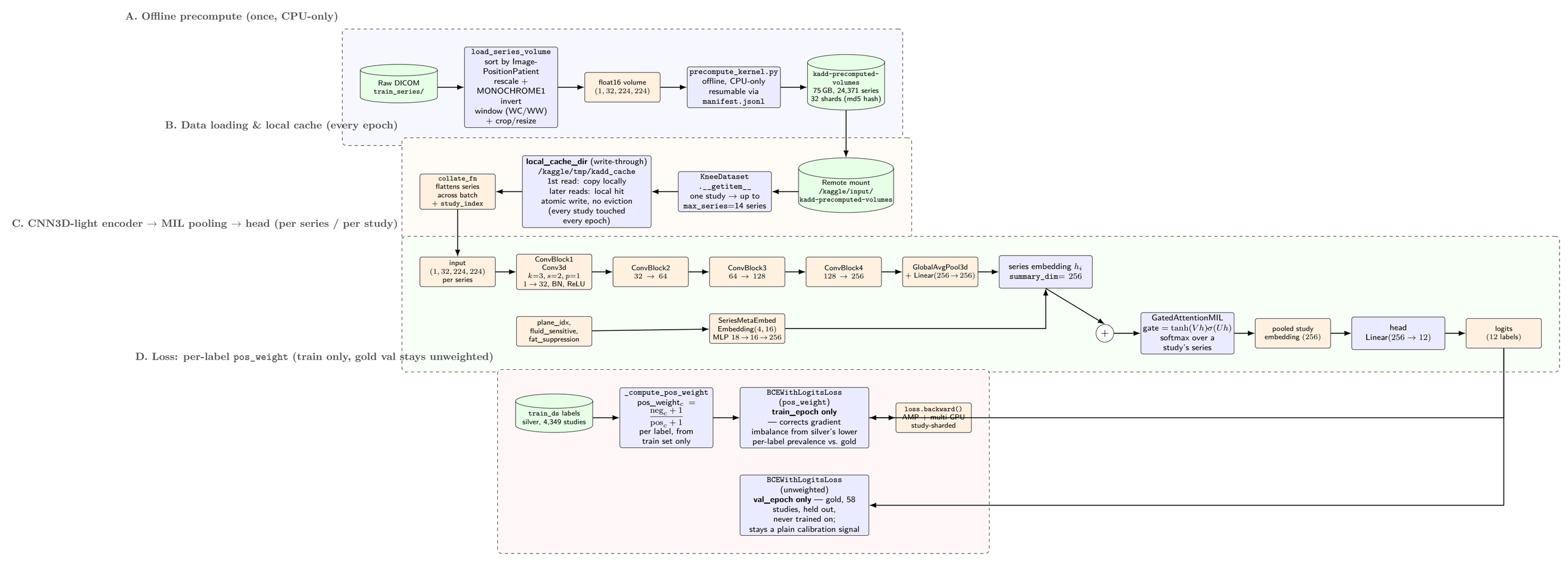
A. Offline precompute (once, CPU-only)
Raw DICOM
train_series/
load_series_volume
sort by Image-PositionPatient
rescale + MONOCHROME1 invert
window (WC/WW) + crop/resize
float16 volume
(1, 32, 224, 224)
precompute_kernel.py
offline, CPU-only
resumable via manifest.jsonl
kadd-precomputed-volumes
75 GB, 24,371 series
32 shards (md5 hash)
B. Data loading & local cache (every epoch)
Remote mount
/kaggle/input/kadd-precomputed-volumes
KneeDataset
.__getitem__
one study → up to max_series=14 series
local_cache_dir (write-through)
/kaggle/tmp/kadd_cache
1st read: copy locally
later reads: local hit
atomic write, no eviction
(every study touched every epoch)
collate_fn
flattens series across batch
+ study_index
C. CNN3D-light encoder → MIL pooling → head (per series / per study)
input
(1, 32, 224, 224)
per series
ConvBlock1
Conv3d
k=3, s=2, p=1
1 → 32, BN, ReLU
ConvBlock2
32 → 64
ConvBlock3
64 → 128
ConvBlock4
128 → 256
GlobalAvgPool3d
+ Linear(256 → 256)
series embedding h_i
summary_dim= 256
plane_idx, fluid_sensitive, fat_suppression
SeriesMetaEmbed
Embedding(4, 16)
MLP 18 → 16 → 256
+
GatedAttentionMIL
gate = $\tanh(Vh)\sigma(Uh)$
softmax over a study's series
pooled study embedding (256)
head
Linear(256 → 12)
logits
(12 labels)
D. Loss: per-label pos_weight (train only, gold val stays unweighted)
train_ds labels
silver, 4,349 studies
_compute_pos_weight
$\text{pos_weight}_c = \frac{\text{neg}_c + 1}{\text{pos}_c + 1}$
per label, from train set only
BCEWithLogitsLoss
(pos_weight)
train_epoch only
— corrects gradient imbalance from silver's lower per-label prevalence vs. gold
loss.backward()
AMP + multi-GPU study-sharded
BCEWithLogitsLoss
(unweighted)
val_epoch only — gold, 58 studies, held out, never trained on; stays a plain calibration signal